

TED (10) 5078

Reg. No.

(Revision - 2010)

Signature.....

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/

TECHNOLOGY - MARCH - 2013

MODEL QUESTION PAPER

MECHATRONICS

(Maximum marks : 100)

[Time : 3 hours]

PART – A

I Answer the following questions in one or two sentences. Each question carries 2 marks.

- 1) Differentiate between incremental and absolute systems
- 2) List out the components of a pneumatic system
- 3) Write the function of a non – return valve
- 4) what is the function of Relays
- 5) what is repeatability ? (5 x 2 = 10)

PART - B

II Answer *any five* of the following. Each question carries 6 marks.

- 1) Explain the mechatronics in manufacturing
- 2) Explain incremental and absolute systems in NC
- 3) With the help of a sketch explain the working of a solenoid valve
- 4) Explain the functions of accumulators and pressure relief valve
- 5) Explain the working of double acting cylinders
- 6) Describe Worm Gear and Bevel Gear
- 7) Distinguish between online and offline programming (5 X 6 = 30)

PART – C

(Answer one full question from each unit. Each question carries 15 marks.)

UNIT – I

III 1. Explain point to point and contouring systems in NC

2. Make comparison between conventional and mechatronics systems 8

OR

- IV 1. Explain the classification of NC systems 7
2. Explain design consideration of NC machine tools 8

UNIT - II

- V a) Draw a pneumatic system and explain the components 7
- b) With the help of a neat sketch explain the working of a Gear pump 8

OR

- VI a) Draw and explain a 4/2 Direction control valve 7
- b) Make a Comparison between Hydraulic and Pneumatic systems 8

UNIT – III

- VII a) Draw & explain single acting and double acting cylinders 8
- b) Explain different types of gears and gear drives 7

OR

- VIII a) Draw a neat sketch of a Vane motor and explain 7
- b) Explain i) Epicyclic gear ii) Rack & pinion 8

UNIT - IV

- IX a) Explain Industrial applications of Robots 8
- b) Describe Lead through programming in Robotics 7

OR

- X a) Explain intelligent robots 7
- b) write short notes on
- i) Tactile sensors ii) Proximity sensors
- iii) Velocity sensors iv) Torque sensors 8

